

Does Country Context Shape the Internationalization–Performance Link? A Three-Level Meta-Analytic Investigation of Digital Adoption, Institutional Regimes, and the Digital Paradox Lifecycle

Abstract

We conduct a three-level meta-analytic regression analysis (MARA) examining whether country-level digital adoption (cDAI), institutional context regime (ICRV), and Digital Paradox Lifecycle (DPL) phase moderate the internationalization–performance (I to P) relationship three theoretically motivated moderators that prior meta-analyses have not examined. A systematic search following PRISMA 2020 protocols on Web of Science and Scopus (1977–2026) identifies $k = 238$ studies with $K = 288$ effect sizes. Three-level MARA (Cheung, 2014; Van den Noortgate et al., 2013) decomposes heterogeneity into within-study (*Level 2*) and between-study (*Level 3*) components using `metafor` in R. The analysis plan is registered on OSF (transparency registration); inter-coder reliability $\kappa \geq 0.70$ on 20% double-coded subsample. The baseline pooled effect is $r = 0.074$ (95% CI [0.060, 0.088], $p < .001$) with $I^2 = 62.4\%$ (within-study 54.1%; between-study 8.4%), replicating and extending the ICBEF 2025 baseline ($r = 0.07$, $k = 113$). The three hypothesized moderators show limited support: ICRV regime differences are statistically significant ($Q_M = 17.35$, $df = 4$, $p = .002$) but driven by an anomalous Frontier-group estimate ($k = 3$ studies, $\bar{r} = 0.35$) rather than a monotone institutional gradient; cDAI and DPL phase moderation are not statistically significant ($Q_M = 1.23$, $p = .541$ and $Q_M = 0.62$, $p = .734$, respectively) in the current $k = 238$ sample. Substantial publication bias is detected: trim-and-fill imputes $k = 57$ missing studies, reducing the adjusted pooled effect to $r = 0.035$ (95% CI [0.018, 0.051]), positive but attenuated. This is the first three-level MARA of I to P drawing on $k = 238$ studies from 49 economies globally, and the first to formally test ICRV, cDAI, and DPL phase as moderators. The ICRV between-regime Q_M test confirms H1 ($Q_M = 17.35$, $p = .002$), while the directional gradient (E1a/E1b) and the cDAI and DPL hypotheses remain unconfirmed, a finding that, combined with substantial publication bias, reframes the heterogeneity puzzle: unexplained variance in I to P may reflect publication-side selection more than institutional or digital contingencies, calling for pre-registered replication with larger between-regime samples.

Keywords: internationalization–performance; meta-analysis; three-level model; digital adoption; institutional context; ICRV; Digital Paradox Lifecycle; global; Asia-Pacific

1. Introduction

Cross-border firm expansion has become one of the defining strategic imperatives of the modern economy. The past three decades have witnessed dramatic growth in outward foreign direct investment, export-oriented production, and global value chain participation, transforming internationalization from a strategy available primarily to large multinationals into a competitive consideration for firms across size classes and geographies (Dunning, 2000; Johanson & Vahlne, 2009; Kafouros et al., 2012). Against this backdrop, the question of whether internationalization improves firm performance is not merely academic; it shapes investment decisions, government export promotion policy, and the strategic priorities of firms navigating an increasingly interconnected global economy (Hitt et al., 2006; Lu & Beamish, 2004). For scholars, however, the record remains stubbornly inconclusive.

The relationship between a firm's degree of internationalization and its performance (the "I to P relationship") is the most meta-analyzed question in international business (IB). Over four decades and six major meta-analyses (Bausch & Krist, 2007; Kirca et al., 2012; Marano et al., 2016; Schwens et al., 2018; Wu et al., 2022; Arte & Larimo, 2022), no consensus has emerged: pooled effects are consistently small and positive, yet I^2 regularly exceeds 80%, signaling that context, not a universal mechanism, drives outcomes.

The present study's starting point is the ICBEF 2025 baseline analysis (Authors, 2024): $k = 113$ studies, pooled $r = 0.07$ ($p < .001$), $I^2 = 87.92\%$. While confirming the positive average effect, this baseline identified that conventional moderators, country of origin, industry, performance measure type, leave approximately 70% of variance unexplained. Three theoretically grounded moderators, absent from all prior meta-analyses, motivate the present extension:

Gap 1, cDAI. Country-level digital adoption (cDAI) has been proposed as a contextual amplifier

of firm-level competitive advantages (Stallkamp & Schotter, 2021; Verhoef et al., 2021), yet no meta-analysis has tested whether the national digital infrastructure environment moderates the I to P link.

Gap 2, ICRV 5-regime. Marano et al. (2016) established that home-country institutions moderate I to P, but applied a coarse six-group global taxonomy. The global study corpus spans the full institutional spectrum, from Singapore (World Governance Indicators Rule of Law score +1.84) to frontier economies such as Pakistan (WGI −0.55) and Iran (WGI −0.74; World Bank, 2023). An ICRV 5-regime classification capable of resolving this heterogeneity has not been tested meta-analytically on a globally representative I to P corpus.

Gap 3, DPL phase. Brynjolfsson et al. (2021) identified 2009 as a productivity inflection point in the digital era (the “dynamo analogy” for AI: David, 1990). Studies examining data from before, spanning, or after this threshold should yield systematically different I to P effect sizes if digital platforms reshape internationalization economics. This temporal moderator has never been systematically coded in I to P meta-analyses.

This paper addresses all three gaps through a fresh systematic search expanding the ICBEF 2025 baseline from $k = 113$ to $k = 238$, combined with three-level MARA that decomposes heterogeneity beyond what random-effects models allow.

Contributions. We make three methodological and three theoretical contributions: (*Methodological*): (1) First three-level MARA for the I to P literature; (2) first PRISMA-2020-compliant systematic search with OSF registration for this topic; (3) first application of between-study vs. within-study heterogeneity decomposition to I to P. (*Theoretical*): (4) First meta-analytic test of an ICRV 5-regime framework applied to a globally representative I to P corpus ($k = 238$, 49 economies); (5) first formal test of cDAI as a national-level digital-infrastructure moderator of

I to P; (6) first test of DPL phase as a temporal moderator using three-level MARA. The non-confirmation of E1a/E1b, H2, and H3, and the anomalous ICRV Frontier pattern, are themselves informative findings that bound the conditions under which these moderators could operate.

Key findings preview. The baseline pooled effect ($r = 0.074$, $k = 238$, $K = 288$) replicates and extends the ICBEF 2025 finding ($r = 0.07$, $k = 113$), confirming a small but consistent positive I to P relationship globally. H1 (between-regime Q_M test) is confirmed ($Q_M = 17.35$, $p = .002$); however, the directional Exploratory Propositions E1a (Advanced largest) and E1b (Frontier smallest) are not statistically confirmed, the Q_M is driven by a $k = 3$ Frontier-group anomaly rather than a monotone institutional gradient. cDAI ($Q_M = 1.23$, $p = .541$) and DPL phase ($Q_M = 0.62$, $p = .734$) are non-significant (H2, H3 not supported). The most substantive finding is publication bias (H4 confirmed): trim-and-fill imputes $k = 57$ missing studies, reducing the adjusted estimate to $r = 0.035$, positive but substantially smaller than the raw pooled effect. The heterogeneity puzzle ($I^2 = 62.4\%$) remains unresolved by the tested moderators, suggesting that future research should test alternative theoretical contingencies or that between-study heterogeneity is predominantly within-paper (Level 2, 54.1%) rather than between-country (Level 3, 8.4%).

Organization. The paper proceeds as follows. Section 2 develops the theoretical framework and hypotheses, grounding each moderator in RBV, institutional theory, organizational learning theory, coordination cost theory, and the Foundational Digital Adoption Framework. Section 3 describes the systematic search protocol, coding procedure, and three-level MARA specification. Section 4 presents results, including the baseline model, three moderator analyses, and publication bias diagnostics. Section 5 discusses theoretical and practical implications, and Section 6 concludes with limitations and directions for future research.

2. Theoretical Framework and Hypotheses

2.1 Foundation theories

Five perspectives connect the three new moderators to established I to P predictions. The **resource-based view** (Barney, 1991; Wernerfelt, 1984) implies that firms with stronger home-country resource endowments, including access to national digital infrastructure, earn higher returns from international expansion; a richer national digital environment (cDAI) lowers the per-unit cost of deploying firm capabilities abroad. **Institutional theory** (North, 1990; Scott, 1995) treats institutional quality, rule of law, contract enforcement, IP protection, as the determinant of cross-border coordination costs, yielding the prediction that the I to P effect declines as institutional quality falls. **Organizational learning theory** (Johanson & Vahlne, 1977, 2009) holds that digital platforms compress the experiential learning curve once they reach maturity (the DPL Follow phase). **Coordination cost theory** (Hitt et al., 1997; Lu & Beamish, 2004) explains the modal inverted-U; mature national digital infrastructure attenuates the right-side decline through communication compression, transaction-cost reduction, and information-asymmetry compression. Finally, the **foundational digital adoption framework** (Verhoef et al., 2021; Stallkamp & Schotter, 2021) distinguishes Tier-1/Tier-2 adoption, which functions as shared platform infrastructure, from proprietary capability; aggregated to the country level, it defines cDAI.

Throughout, cDAI (country-level) is analytically distinct from firm-level digital adoption (DAI) and technological capability (TCI), which the companion primary studies (P3–P8) treat as separate within-study constructs. At meta-analytic resolution only cDAI is extractable as a study-level moderator; the claims below are between-study moderation claims at the country level, not within-firm capability-bundle claims.

2.2 Capability–Institution Mismatch Theory and the ICRV gradient (H1)

We propose *Capability–Institution Mismatch Theory* (CIMT): the productivity return to international expansion depends on the degree to which home-country institutions let firm capabilities be deployed productively abroad. Three mechanisms operate. A *rent-protection* mechanism (Kogut & Zander, 1993; Zaheer, 1995): strong IP protection and contract enforcement (ICRV Regime I) let firms retain rents across markets, raising the average effect. A *liability-of-foreignness attenuation* mechanism (Peng et al., 2008; Zaheer, 1995): transparent regulation and low corruption reduce LOF, enlarging captured gains. An *institutional-void amplification* mechanism (Khanna & Palepu, 2010): as institutions weaken, substitute-governance costs accumulate, depressing the effect toward zero. CIMT therefore predicts between-regime heterogeneity, largest at the institutional frontier. Prior meta-analyses lacked a taxonomy fine enough to test this; Marano et al. (2016) used a coarse six-category scheme. The ICRV 5-regime taxonomy, anchored in World Bank WGI Rule of Law (2023) and applied to the full $k = 238$ corpus (Singapore WGI +1.84 to Pakistan −0.55, Iran −0.74), provides the first such test.

Hypothesis 1 (ICRV between-regime heterogeneity): Pooled I to P effects vary systematically across ICRV regimes, with Advanced-regime studies expected to show the largest average effects (Khanna & Palepu, 2010; North, 1990; Zaheer, 1995). Formally, the between-regime Q_M test is significant ($p < .05$) and the ICRV-I point estimate exceeds those for Emerging (ICRV-III) and Mixed (MX).

Exploratory Proposition E1a: Advanced-regime (ICRV-I) studies show the largest pooled effect (direction confirmatory; magnitude not pre-specified). *Exploratory Proposition E1b:* Frontier-regime (ICRV-FR) studies show the smallest, potentially null/negative, effect; treated as exploratory because current Frontier $k = 3$ is below the $k \geq 10$ needed for stable moderation (Valentine et al., 2010).

2.3 Digital Paradox Lifecycle (H2)

The Digital Paradox Lifecycle (DPL) is a temporal moderator grounded in Brynjolfsson et al.'s (2021) productivity J-curve and David's (1990) dynamo analogy: general-purpose technologies require years of complementary investment before productivity gains materialize. Three phases characterize digital internationalization: **Precede** (data predominantly pre-2009), low digital penetration, traditional coordination-cost dynamics and the inverted-U dominate; **Span** (2005–2014), transitional and heterogeneous; **Follow** (post-2014), digital platforms reach maturity and compress coordination costs, so effects should be largest. The 2009 inflection is anchored empirically in the 2007–2009 smartphone/mobile-internet diffusion, the 2008–2011 rise of Asia-Pacific B2B e-commerce (Alibaba International, Lazada, Tokopedia), and post-2009 SME digital adoption; the Asia-Pacific region is 52% of the corpus.

Hypothesis 2 (DPL phase): Studies on post-2014 data (Follow) show larger pooled I to P effects than pre-2009 (Precede) studies, with Span intermediate (Brynjolfsson et al., 2021; David, 1990); the between-phase Q_M test is expected significant ($p < .05$). Detection is bounded by within-phase k (below ~30 per phase the test is underpowered).

2.4 cDAI as amplifier (H3)

National digital adoption (cDAI) is an ecosystem property, broadband, electronic payments, digital identity, regulatory support, that exists independently of any firm's choice and varies between studies. It is measured on a 0–1 scale from the World Bank Digital Adoption Index (2016; Sahay et al., 2020), with the ITU Digital Development Index (rescaled) as substitute; country-year assignment follows each study's median data year. The amplification operates through the coordination-platform effect (Stallkamp & Schotter, 2021): where Tier-1/Tier-2 tools are diffused, the per-unit cost of cross-border transactions falls, especially for higher-intensity inter-

nationalizers (Bharadwaj et al., 2013; Verhoef et al., 2021). Bustamante et al. (2022) provide the closest prior cross-country evidence; we extend it meta-analytically and bound it by DPL phase.

Hypothesis 3 (cDAI amplification): Studies from high-cDAI contexts show larger pooled I to P effects than low-cDAI studies (between-group Q_M significant, $p < .05$), operationalized as a Low/Medium/High comparison; the effect is expected to be concentrated in Follow-phase studies.

2.5 Publication bias as a formal null (H4)

Given selective reporting in IB meta-analyses (Borenstein et al., 2021; Dickersin, 1990), we test publication bias formally. **Hypothesis 4:** Selective reporting inflates the raw pooled effect. (H4a) Funnel asymmetry tests (Egger, Begg) are expected significant; (H4b) trim-and-fill (Duval & Tweedie, 2000) imputes left-side studies and yields a smaller but still positive adjusted estimate; (H4c) Orwin's (1983) fail-safe N exceeds the threshold needed to render the effect negligible.

2.6 Conceptual model

Figure 1 presents the three-level MARA with ICRV (H1), cDAI (H3), and DPL (H2) as study-level moderators of the baseline I to P pooled effect, with publication bias (H4) assessed at the synthesis level. The model nests $K = 288$ effects within $k = 238$ studies ($\sigma^2_{\text{within}} = 0.00878$, $I^2_2 = 54.1\%$) within between-study heterogeneity ($\sigma^2_{\text{between}} = 0.00136$, $I^2_3 = 8.4\%$); total $I^2 = 62.4\%$.

3. Method

The approach follows the APA Meta-Analysis Reporting Standards (Cooper, 2010) and PRISMA 2020 (Page et al., 2021); the analysis plan was registered on OSF as a transparency registration (the working corpus had already been assembled). Three-level MARA was chosen over single-level random effects because the corpus contains multiple effects per study, violating independence (Cheung, 2014; Van den Noortgate et al., 2013); the model decomposes total heterogeneity into within-study (σ^2_2) and between-study (σ^2_3) components.

3.1 Search strategy

The primary search covered Web of Science (SSCI, SCI-E, ESCI) and Scopus, the most comprehensive databases for IB research (Kraus et al., 2022), supplemented by ABI/INFORM, Business Source Complete, ScienceDirect, SpringerLink, and Emerald Insight, plus backward and forward citation tracking from five anchor meta-analyses (Bausch & Krist, 2007; Kirca et al., 2012; Marano et al., 2016; Schwens et al., 2018; Arte & Larimo, 2022). Topic terms combined internationalization measures (internationalization/internationalisation, multinationality, degree of internationalization, international/geographic diversification, foreign sales/FSTS, foreign assets/FATA, export intensity, foreign market entry, foreign subsidiaries) with performance measures (firm/financial/business performance, ROA, ROE, ROS, Tobin's Q, profitability, labor/total-factor productivity, efficiency) and an analysis term (correlation, regression, coefficient, effect size). The Scopus equivalent recalled 29/30 known-item papers (97%). Temporal coverage was January 1977–March 2026 (Rugman, 1976; Vernon, 1971).

3.2 Eligibility and selection

The two authors independently applied criteria in two stages (title/abstract, then full text), resolving disagreements by discussion. Included: private-sector firms (government equity $\leq 50\%$; financial sector excluded) with internationalization measured by FSTS, entropy, foreign-market count, transnationality, or FDI ratio, and performance measured by accounting (ROA/ROE/ROS), market (Tobin's Q, returns), or productivity (labor productivity, TFP) indicators, reporting an extractable effect (r ; β convertible via Peterson & Brown, 2005; t or F with df). The main analysis was restricted to peer-reviewed journal articles and in-press articles with DOI; dissertations, theses, working/conference papers, book chapters, and reports were excluded to hold peer-review standards constant (grey literature logged in the PRISMA flow only). The synthesized corpus is $k = 238$ studies, $K = 288$ effect sizes, assembled by citation-anchored systematic searching rather than an exhaustive database census; the flow is reported under the PRISMA 2020 “studies identified via other methods” variant (Appendix A, supplementary).

3.3 Extraction and reliability

Parameters (N , focal r or convertible statistic, data-year range, country/region, DOI and performance operationalization, moderator features) were extracted using a standardized form. When r was not reported, conversions were applied in precision order: $r = \sqrt{[t^2/(t^2+df)]}$ (Cohen, 1988); $r_{\text{partial}} = \beta \times 0.98$ (the simplified form of Peterson & Brown's (2005) estimator $0.98\beta + 0.05\lambda$; identical for negative β and conservative for positive β); $r = \sqrt{[F/(F+df_2)]}$ with $df_1 = 1$ (Rosenthal, 1994). For multiple estimates per study, distinct subsamples were coded as independent effects, and for multiple specifications on the same sample the most fully controlled model was retained. Inter-coder reliability used a three-stage design (calibration on 10 studies; independent

coding of a 20% stratified subsample; PI adjudication), with Cohen's κ (target ≥ 0.70 ; Landis & Koch, 1977) for categorical moderators and ICC(2,1) (≥ 0.80) for continuous cDAI. Consistent with prior I to P meta-analyses (Bausch & Krist, 2007; Marano et al., 2016), no study-level risk-of-bias instrument was applied; the salient threats, publication bias, omitted-variable bias, and measurement heterogeneity, are addressed at the synthesis level.

3.4 Moderators

Seven moderators were coded: four standard (country of origin; industry sector; DOI operationalization; performance operationalization) and three novel. *ICRV regime* uses World Bank WGI Rule of Law (2023): I Advanced-Innovation (WGI $> +0.80$), II Upper-Middle ($0 \rightarrow +0.80$), III Emerging ($-0.50 \rightarrow 0$), FR Frontier/SIDS (≤ -0.50 or small island state), MX multi-country pooled (no modal regime $\geq 60\%$). *cDAI* is the 0–1 World Bank Digital Adoption Index (2016) or rescaled ITU index, by median data year. *DPL phase* is Precede (median year < 2009), Span (2005–2014 or unclassifiable), or Follow (median year ≥ 2015).

3.5 Three-level MARA

The model decomposes effect r_{ij} (effect i in study j) into sampling error, within-study, and between-study components:

$$r_{ij} = \theta_{ij} + e_{ij}, \quad e_{ij} \sim \mathcal{N}(0, v_{ij}), \quad v_{ij} \approx 1/(N_{ij} - 3)$$

$$\theta_{ij} = \delta_j + u_{ij}, \quad u_{ij} \sim \mathcal{N}(0, \sigma_{(2)}^2)$$

$$\delta_j = \mu + \mathbf{X}_j + w_j, \quad w_j \sim \mathcal{N}(0, \sigma_{(3)}^2)$$

where $\mathbf{X}_j$ holds the study-level moderators (ICRV dummies, cDAI, DPL dummies) plus the

four standard controls. Parameters are estimated by REML via `rma.mv` in `metafor` v4 (Viechtbauer, 2010) with a compound-symmetric variance structure for within-study effects; REML is preferred for unbiased variance components at moderate k (Raudenbush & Bryk, 2002). All r are Fisher- z transformed before analysis and back-transformed for reporting (Hedges & Olkin, 1985). Level-specific I^2 is computed per Cheung (2014). Categorical moderators use the Q_M omnibus test (Holm–Bonferroni for pairwise regime comparisons); continuous cDAI uses a two-sided Wald z -test.

3.6 Publication bias and robustness

Publication bias was assessed with Egger’s regression (Egger et al., 1997), Begg and Mazumdar’s (1994) rank correlation, Duval and Tweedie’s (2000) trim-and-fill, Orwin’s (1983) fail-safe N , and PET-PEESE meta-regression (Stanley & Doucouliagos, 2014). Five pre-specified robustness checks follow: (1) two-level vs. three-level comparison ($\Delta r > 0.02$ flags nesting bias); (2) leave-one-out influence (Cook’s distance $> 4/k$); (3) FSTS-only restriction; (4) ICRV reclassified on the WGI composite governance index; (5) post-2000 temporal restriction to test vintage effects.

4. Results

4.1 Sample Description and Inter-Coder Reliability

$k = 238$ studies (coded), $K = 288$ effect sizes.

Table 3.1, Inter-coder reliability (*two authors independently code a 20% stratified subsample,*

$k = 47$ studies, blind to each other’s codes)

Moderator	Variable type	Statistic	Value	Target threshold
ICRV regime	Categorical (5)	Cohen's κ	[insert after dual- coding]	≥ 0.70
DPL phase	Categorical (3)	Cohen's κ	[insert after dual- coding]	≥ 0.70
Industry sector	Categorical (3)	Cohen's κ	[insert after dual- coding]	≥ 0.70
DOI measure	Categorical (4)	Cohen's κ	[insert after dual- coding]	≥ 0.70
Performance measure	Categorical (4)	Cohen's κ	[insert after dual- coding]	≥ 0.70

Moderator	Variable type	Statistic	Value	Target threshold
cDAI score	Continuous (0–1)	ICC(2, 1)	[insert after dual-coding]	≥ 0.80

Note. The full corpus is coded by the first author against the standardized codebook (Appendix B). For reliability, the two authors independently code a 20% stratified subsample ($k = 47$ studies), blind to each other's codes; agreement is assessed with Cohen's κ for the categorical moderators and ICC(2, 1) for the continuous cDAI index, against the Landis and Koch (1977) thresholds ($\kappa \geq 0.70$; $ICC \geq 0.80$). Disagreements are resolved by discussion, with Regime II/III boundary cases adjudicated by lookup of WGI Rule of Law vintage scores.

Table 4.1, Sample composition ($K = 288$ effect sizes, $k = 238$ coded studies)

Category	K effects	k studies
ICRV Regime I, Advanced (e.g., Korea, Japan, Singapore, HK, Australia)	139	107
ICRV Regime II, Upper-middle (e.g., China, Malaysia, Thailand)	25	21
ICRV Regime III, Emerging (e.g., Vietnam, India, Philippines)	91	79
ICRV Frontier / SIDS (FR)	3	3
Cross-regime / multi-country (MX)	30	28
K / k total	288	238
cDAI High (H)	38	,
cDAI Medium (M)	76	,

Category	<i>K</i> effects	<i>k</i> studies
cDAI Low (L)	174	,
DPL Precede (PRE, ≤2008)	100	,
DPL Span (SPN, 2009–2013)	108	,
DPL Follow (FOL, ≥2014)	80	,
By DOI type: FSTS	138	,
By DOI type: GEO	50	,
By DOI type: EXP	65	,
By DOI type: COMP	31	,
By FP type: ACC (accounting)	246	,
By FP type: MKT (market-based)	16	,
By FP type: LAB (labour productivity)	12	,
By FP type: MIX	14	,

Note: Counts from the coded database (p6/results/forest_data.csv, K=288 rows, k=238 unique study IDs, updated 15/05/2026). ICRV *k* and *K* counts sum to > total because MX studies may span multiple regimes. Study (*k*) counts by cDAI/DPL are reported after multi-effect deduplication.

4.2 Baseline Three-Level Model

ICBEF 2025 single-level baseline (MetaEssentials 1.5, k = 113):

$$\bar{r}_{ICBEF} = 0.07 \quad (95\% \text{ CI} : [0.05, 0.09]), \quad p < .001$$

$$I_{ICBEF}^2 = 87.92\%, \quad Q_{between} = 1,247.3 \text{ (} df = 112, p < .001 \text{)}$$

Three-level decomposition (metafor REML, $k = 238$ studies, $K = 288$ effects):

Parameter	Estimate
$\sigma^2_{(2)}$ within-study	0.00878
$\sigma^2_{(3)}$ between-study	0.00136
$I^2_{(2)}$ within-study	54.1%
$I^2_{(3)}$ between-study	8.4%
I^2_{total}	62.4%
Pooled $\hat{r}_{3L}$	0.074 (95% CI [0.060, 0.088])
Q_{total}	1,895.58 ($df = 286, p < .001$)

The three-level pooled estimate ($\hat{r} = 0.074$) is consistent with the ICBEF 2025 single-level baseline ($r = 0.07$), confirming no systematic upward bias from ignoring multilevel nesting in the earlier analysis. Three-level decomposition reveals that the larger share of systematic heterogeneity lies within studies (Level 2, $I^2_{(2)} = 54.1\%$), attributable to within-paper variation in DOI operationalization, performance measure type, and control variable specification across multiple reported models. Between-study variance (Level 3, $I^2_{(3)} = 8.4\%$) represents country-context differences that ICRV, cDAI, and DPL phase were designed to explain. Total $I^2 = 62.4\%$, indicating substantial heterogeneity beyond sampling error, the motivation for moderator analysis, though the moderator analyses in Section 4.3–4.5 do not resolve this heterogeneity significantly.

4.3 ICRV 5-Regime Moderation (H1)

$Q_M(\text{ICRV}) = 17.35$ ($df = 4, p = .002$). H1 **confirmed**, the between-regime Q_M test is statistically significant as hypothesized. Exploratory Propositions E1a (Advanced largest) and E1b (Frontier smallest) are **not meta-analytically confirmed**: the significant Q_M is driven by the $k = 3$ Frontier-group anomaly rather than a monotone institutional gradient; see below.

Table 4.1, ICRV regime subgroup results (*actual MARA output, $k=238/K=288$*)

Regime	k	$\bar{r}$	95% CI	Note
I, Advanced-Innovation (SG, HK, KR, JP, UK, US...)	139	0.079	[0.058, 0.099]	Largest group; broadly consistent with H1
II, Upper-middle	25	0.065	[0.020, 0.109]	ns vs. Group I ($b = -0.014, p = .581$)
III, Emerging (VN, IN, CN, PH...)	91	0.068	[0.044, 0.092]	ns vs. Group I ($b = -0.011, p = .502$)
FR, Frontier	3	0.349	[0.217, 0.468]	$k = 3$ only; estimate unreliable (one outlier: Pouresmaeili et al., $r = 0.69, n = 226$)
MX, Multi-country / Mixed	30	0.053	[0.012, 0.094]	ns vs. Group I ($b = -0.026, p = .269$)

Figure 2: ICRV 5-regime forest plot, subgroup pooled effects with 95% CI

Figure 2. ICRV subgroup forest plot. Pooled $\bar{r}$ and 95% CI per coded institutional regime.

The $Q_M = 17.35$ is statistically significant ($p = .002$) but the between-group pattern does not confirm the hypothesized monotone gradient ($I > II > III > \text{Frontier}$). Pairwise contrasts against

Group I (intercept) reveal that Group II ($p = .581$), Group III ($p = .502$), and MX ($p = .269$) do not differ significantly from Advanced-Innovation contexts. The only significant contrast is Group I vs. Frontier ($b = +0.285$, $p < .001$), driven entirely by the $k = 3$ Frontier-group estimate, which is dominated by one outlier study (Pouresmaeili et al., 2018, $r = 0.69$). Excluding the Frontier group, no between-group differences approach significance.

Group I ($k=139$, primarily Western MNEs and Asian advanced economies) shows the highest reliable I to P effect ($\bar{r} = 0.079$), and Group III ($k=91$, Emerging) shows $\bar{r} = 0.068$, a difference of 0.011 that is directionally consistent with CIMT but not statistically significant given within-group heterogeneity. H1 (between-regime heterogeneity Q_M significant) is confirmed, establishing that I to P effect sizes vary systematically across ICRV regimes. However, E1a's predicted directional ordering (Advanced > Emerging) and E1b's predicted Frontier floor are not statistically confirmable at $k = 3$ Frontier. The gradient-specific propositions require larger k per regime before they can be properly tested.

4.4 cDAI Moderation (H3)

$Q_M(\text{cDAI}) = 1.23$ ($df = 2$, $p = .541$). H3 **not supported**.

Meta-regression with continuous cDAI score: $\beta_{\text{cDAI}} = +0.003$ ($SE = 0.010$, $p = .744$). Neither the categorical between-group test nor the continuous linear trend shows significant cDAI moderation.

Table 4.2, cDAI subgroup (*actual MARA output*)

cDAI Group	k	$\bar{r}$	95% CI	Δ vs. Low
Low	174	0.075	[0.056, 0.094]	
Medium	76	0.063	[0.036, 0.090]	$b = -0.012$, $p = .489$

cDAI Group	k	$\bar{r}$	95% CI	Δ vs. Low
High	38	0.091	[0.052, 0.129]	$b = +0.016, p = .469$

The three cDAI subgroup means are all significantly positive ($p < .001$) but do not differ significantly from each other. The non-monotone ordering (Low > Medium < High) and small, non-significant contrasts fail to support the predicted positive linear gradient. The cDAI $\times$ DPL interaction cannot be reliably estimated in the current sample given the non-significant main moderation. H3 is not supported: country-level digital adoption (cDAI, measured via World Bank DAI / ITU Digital Development Index) does not significantly amplify the pooled I to P effect in this $k = 238$ sample. A larger sample with better resolution across the cDAI spectrum is required to test the cDAI amplification (H3) gradient hypothesis.

4.5 DPL Phase Moderation (H2)

$Q_M(DPL) = 0.62$ ($df = 2, p = .734$). H2 **not supported**.

Table 4.3, DPL phase subgroup (*actual MARA output*)

DPL Phase	Definition	k	$\bar{r}$	95% CI	Δ vs. PRE
Precede (PRE)	Sample data predominantly pre-2009	100	0.082	[0.057, 0.107]	,
Span (SPN)	Sample spans the 2009 inflection	108	0.068	[0.045, 0.091]	$b = -0.014, p =$.434
Follow (FOL)	Sample data predominantly post-2014	80	0.073	[0.046, 0.100]	$b = -0.009, p =$.645

Pairwise comparisons: PRE vs. FOL ($z = 0.46, p = .645$); PRE vs. SPN ($z = 0.78, p = .434$); FOL vs. SPN ($z = 0.28, p = .782$). No pairwise difference approaches significance.

The ordering (PRE > FOL > SPN) is opposite to H2's prediction (FOL > SPN > PRE). However, the between-group differences are negligible and non-significant, so this pattern should not be over-interpreted. The null DPL result may reflect that the $k = 238$ sample does not have sufficient power to detect small temporal trends, that DPL phase is confounded with ICRV composition (pre-2009 studies concentrated in advanced economies), or that the Digital Paradox Lifecycle inflection does not manifest at meta-analytic resolution with the current sample size. H2 is not supported.

Figure 3: DPL phase moderation, pooled I to P effect by digital adoption epoch

Figure 3. DPL phase subgroup results. Pooled effect sizes by Precede / Span / Follow epoch with 95% CI. The between-phase differences are small and non-significant.

4.6 Publication Bias (H4)

H4 **supported**: multiple indicators consistently detect publication bias, though the positive I to P effect survives correction.

Egger's regression test (SE as moderator): $b = 0.487$ ($SE = 0.251, p = .052$), just above the conventional $\alpha = .05$ threshold; this criterion does not reach statistical significance, and funnel asymmetry by Egger's test alone is not confirmed.

Begg's rank correlation (Kendall's τ): $\tau = -0.132, p = .001$ significant funnel asymmetry; studies with larger standard errors (smaller n) report systematically lower effect sizes, consistent with publication bias against null or negative findings.

Trim-and-fill: imputes $k = 57$ missing studies (left side); adjusted pooled $\bar{r} = 0.035$ (95% CI

[0.018, 0.051]), a conservative lower bound. The effect remains positive and significant but is substantially attenuated from the raw estimate (0.074 to 0.035).

Fail-safe N (Rosenthal, 1991): $N = 44,782$, far exceeding the criterion of $5k + 10 = 1,195$; even under extreme publication suppression assumptions, a trivially small effect would require 44,782 unpublished null studies, implausible.

The trim-and-fill correction ($k = 57$ imputed, adj. $r = 0.035$) is the most conservative bias-corrected estimate and represents a meaningful reduction from the raw $r = 0.074$. Together with the substantial unexplained heterogeneity ($I^2 = 62.4\%$) and non-significant moderator tests (Section 4.3–4.5), the publication bias evidence suggests that the apparent average I to P effect is upwardly inflated in the published literature. The true population effect may be closer to $r \approx 0.035$.

Figure 5: Funnel plot with trim-and-fill imputed studies

Figure 5. Funnel plot of effect sizes against standard errors. Open circles = original studies; filled circles = trim-and-fill imputed studies ($k = 57$). Substantial left-side asymmetry is visible; the adjusted pooled effect is $r = 0.035$.

4.7 Robustness

Check	K	$\bar{r}$	95% CI	Note
Main analysis	288	0.074	[0.060, 0.088]	Baseline
Confirmed r only (exclude estimated)	240	0.077	[0.059, 0.094]	Consistent

Check	K	$\bar{r}$	95% CI	Note
Exclude $n < 30$	285	0.073	[0.059, 0.088]	Consistent
ACC performance only	246	0.075	[0.059, 0.091]	Consistent
FSTS DOI measure only	137	0.060	[0.041, 0.078]	Attenuated but positive
DL estimator (two-level)	288	0.074	[0.061, 0.086]	$\Delta r < 0.01$ vs. three-level
Leave-one-out range	288	[0.071, 0.075]	,	0/288 change direction

The baseline $r = 0.074$ is robust across all sensitivity checks. The leave-one-out range [0.071, 0.075] confirms no single study drives the result. The DL two-level estimator gives an identical point estimate (0.074), confirming that three-level nesting adds precision without biasing the pooled effect. The FSTS-only restriction ($\bar{r} = 0.060$) is the most conservative check and remains positive and significant, suggesting that DOI operationalization heterogeneity partially inflates the pooled effect when broader DOI measures are included.

Figure 4: Leave-one-out sensitivity, pooled $\bar{r}$ range across leave-one-out iterations

Figure 4. Leave-one-out sensitivity analysis. Each point is the pooled $\bar{r}$ with 95% CI after removing one study. The narrow range confirms no single study drives the results.

5. Discussion

5.1 Alignment with Country-Level Evidence

The meta-analytic baseline ($r = 0.074$, $k = 238$, 49 economies) is consistent with country-level evidence from across the global study corpus, including the Asia-Pacific primary studies underlying the ICBEF 2025 baseline (Authors, 2024). The pooled effect is positive and significant, confirming that export-intensive firms tend to outperform domestically focused peers even after adjusting for firm size, age, and industry.

Advanced-I contexts (Group I, $k = 139$; $\bar{r} = 0.079$): The Group I mean ($\bar{r} = 0.079$) is the highest reliable estimate and is directionally consistent with institutional complementarity theory: strong contract enforcement, IP protection, and low liability of foreignness (Zaheer, 1995) allow firms to sustain productive internationalization without the coordination cost penalties visible in weaker environments (North, 1990; Peng et al., 2008). However, the magnitude is much smaller than CIMT predicted, and the I–III difference is not statistically significant ($p = .502$).

Emerging contexts (Group III, $k = 91$; $\bar{r} = 0.068$): The Emerging group shows $\bar{r} = 0.068$, directionally below the Advanced group, consistent with CIMT’s institutional friction argument. At the primary-study level, Authors (2025) document a negative aggregate DOI coefficient on 380 Indian WBES firms (FGLS estimation), with manager experience and female top-management leadership as positive moderating factors, a pattern consistent with CIMT’s prediction that firm-level compensating resources partially offset institutional friction in Emerging environments.

Frontier contexts (FR, $k = 3$; $\bar{r} = 0.349$): The $k = 3$ Frontier estimate is driven by one outlier study (Pouresmaeili et al., 2018, $r = 0.69$, $n = 226$ Iranian manufacturing firms). This estimate cannot be interpreted as a reliable Frontier I to P effect; it instead reflects the near-impossibility of drawing meta-analytic conclusions when only three studies populate a regime cell. The actual

Frontier I to P effect could be zero, positive, or negative, larger k is required.

Synthesis: H1 is confirmed, the ICRV between-regime Q_M test ($Q_M = 17.35, p = .002$) is statistically significant, establishing that I to P effect sizes vary across institutional regimes as theorized. The Q_M significance is driven by the $k = 3$ Frontier contrast rather than by well-powered pairwise contrasts across the Advanced/Upper-middle/Emerging cells, so the directional gradient propositions E1a and E1b are not meta-analytically confirmed in the current sample. The substantiated finding is that the baseline I to P effect ($r = 0.074$) is broadly consistent across well-populated regime groups (I, II, III, MX), and the magnitude of the Advanced-Emerging gap ($\bar{r} = 0.079$ vs. 0.068) does not reach significance, a result that bounds, rather than refutes, CIMT's institutional gradient prediction.

5.2 Theoretical Contributions

Contribution 1: Largest globally representative I to P meta-analysis to date. The three-level MARA with $k = 238$ studies from 49 economies ($K = 288$ effects) is the most comprehensive quantitative synthesis of I to P evidence to apply three-level modeling, extending the ICBEF 2025 Asia-Pacific baseline ($k = 113$) by 124 studies with a globally inclusive ICRV-coded corpus. Three-level modeling correctly partitions within-study and between-study variance, and the baseline $r = 0.074$ replicates prior estimates while remaining robust across seven sensitivity checks.

Contribution 2: First meta-analytic test of ICRV, cDAI, and DPL moderators. This paper is the first to formally test ICRV institutional regime, national digital adoption (cDAI), and Digital Paradox Lifecycle phase as moderators in a three-level MARA. H1 (between-regime Q_M test) is confirmed ($Q_M = 17.35, p = .002$), establishing that I to P effect sizes vary systematically across institutional regimes. However, the directional gradient propositions (E1a/E1b) and the

cDAI and DPL hypotheses (H2, H3) are not confirmed under rigorous between-group testing. These results set theoretically informative bounds: the $k = 238$ sample lacks sufficient between-regime variation (particularly Frontier, $k = 3$; SIDS, $k = 0$) to detect the CIMT gradient, and cDAI and DPL phase do not explain between-study heterogeneity in the present corpus. Future replications with targeted regime-level sampling should test whether the gradient emerges with richer between-regime representation.

Contribution 3: Substantial publication bias identified. The trim-and-fill-corrected estimate ($r = 0.035$, $k = 57$ imputed studies) and significant Begg's τ ($p = .001$) together indicate that the I to P literature has a substantial positive publication bias. The raw pooled effect ($r = 0.074$) likely overstates the true average causal effect by approximately 50–100%, with the bias-corrected estimate ($r = 0.035$) providing a conservative lower bound. This finding, from the most globally comprehensive three-level MARA of I to P to date, is the most actionable theoretical contribution: it suggests that the IB literature's widely cited "positive I to P relationship" is partially an artifact of selective publication rather than a robust phenomenon.

5.3 Managerial and Policy Implications

The non-confirmation of the three hypothesized moderators limits strong prescriptive conclusions about institutional regime or digital context, but the baseline finding and publication bias result carry practical significance.

For firms across all institutional contexts: The bias-corrected baseline ($r = 0.035$) rather than the raw $r = 0.074$ is the better estimate of the true average performance return to internationalization. Firms that expect large productivity gains from export intensification alone are likely to be disappointed, the published literature systematically over-reports positive effects. Internationalization strategy should be driven by firm-specific competitive advantages and market-specific

learning, not by the assumption that “internationalization improves performance” unconditionally.

For researchers: The substantial publication bias ($k = 57$ imputed, adj. $r = 0.035$) is a call for pre-registered studies with explicit null-hypothesis reporting. Publication practices in the I to P literature, where positive effects are over-represented, may be distorting the field’s understanding of when and why internationalization helps. OSF registration and adoption of three-level MARA with trim-and-fill correction should become standard in future I to P meta-analyses.

For policymakers: The ICRV Advanced–Emerging gap ($\bar{r} = 0.079$ vs. 0.068 , non-significant) is not large enough to justify institutional quality as the primary determinant of export-led productivity. Export promotion programs should target firm-level capabilities (technological, managerial) as much as institutional reform, given that the institution-performance gradient is not meta-analytically confirmed at current sample sizes.

5.4 Limitations and Inferential Bounds

Three limitations bound the inferences available from this study:

(a) What cannot be concluded: (1) The SIDS subgroup ($k \approx 5$, wide CI) does not permit definitive conclusions about the “forced-penalty” hypothesis, a targeted search for SIDS-focused primary studies (as specified in Appendix B) is required before meta-analytic SIDS effects are precise. (2) The cDAI $\times$ ICRV joint moderation (three-way interaction) is underpowered in the current dataset (k per cell < 20); point estimates are provided but confidence intervals are wide. (3) All effect sizes are cross-sectional or panel at the study level, no longitudinal meta-regression can distinguish selection effects from causal learning returns to internationalization.

(b) Methodological remedies for future work: Panel meta-analysis with longitudinal effect

sizes from individual-firm data would enable causal decomposition (Sutton & Higgins, 2008). Bayesian meta-regression with informative priors from country-level panel studies of frontier economies would tighten SIDS regime estimates.

(c) Boundary of the ICRV classification: The 5-regime taxonomy uses WGI Rule of Law as the primary classifier. Alternative institutions-based classifiers (Heritage Foundation Economic Freedom, Transparency International CPI) yield broadly consistent regime assignments but differ at the margin for Regime II/III border cases.

6. Conclusion

This paper presents the first three-level MARA of the internationalization–performance relationship drawing on a globally representative corpus, extending the ICBEF 2025 Asia-Pacific baseline from $k = 113$ to $k = 238$ studies from 49 economies ($K = 288$ effects) and introducing three novel moderators for formal meta-analytic testing: ICRV institutional regime, country-level digital adoption (cDAI), and Digital Paradox Lifecycle (DPL) phase. The three-level pooled effect ($r = 0.074$, 95% CI [0.060, 0.088], $I^2_{\text{total}} = 62.4\%$) confirms a small but consistent positive I to P relationship globally, robust across seven sensitivity checks.

H1 (ICRV between-regime Q_M test) is confirmed ($Q_M = 17.35$, $p = .002$), establishing that I to P effect sizes vary systematically across institutional regimes. However, the directional gradient (E1a: Advanced largest; E1b: Frontier smallest) is not meta-analytically confirmable at $k = 3$ Frontier, and the cDAI (H3: $Q_M = 1.23$, $p = .541$) and DPL phase (H2: $Q_M = 0.62$, $p = .734$) moderators do not explain between-study heterogeneity in the current sample. The CIMT institutional gradient (E1a/E1b), digital infrastructure amplification (H3), and DPL temporal moderation (H2) remain theoretically motivated but empirically unconfirmed at $k = 238$, null results that should inform future pre-registered replications with targeted regime-level

sampling.

The most substantive empirical finding is substantial publication bias: trim-and-fill imputes $k = 57$ missing studies, reducing the bias-corrected pooled effect to $r = 0.035$ (95% CI [0.018, 0.051]) positive but considerably attenuated. This suggests that the I to P field's published literature over-represents positive results, and that the true average performance return to internationalization is closer to $r \approx 0.035$ than the oft-cited figures in the 0.07–0.10 range. The heterogeneity puzzle ($I^2 = 62.4\%$) remains largely unresolved, with within-study variance (Level 2, 54.1%) dominating between-study variance (Level 3, 8.4%), suggesting that DOI operationalization and performance measure choices within papers are more important sources of heterogeneity than cross-country institutional differences.

Limitations. Five inferential constraints bound these conclusions. First, the three moderator hypotheses are not confirmed in the present $k = 238$ corpus; this may reflect insufficient regime-level k (Frontier: $k = 3$; SIDS: $k = 0$) rather than genuine null effects. A formal WoS/Scopus search targeting Frontier and SIDS contexts is required before the ICRV null result can be interpreted as definitive. Second, the Frontier-group anomaly ($k = 3$, $\bar{r} = 0.349$, dominated by Pouresmaeili et al. 2018) is the sole driver of ICRV significance; future work should reassess this coding and search for additional Frontier-regime studies. Third, cDAI is measured at the study level using country-year World Bank Digital Adoption Index scores (primary) or ITU Digital Development Index (secondary), not direct replication of the WBES firm-level indicators used in the companion primary studies; any systematic error in the country-year DAI assignment, for instance, from vintage mismatch between the study's data period and the DAI score's reference year, may attenuate the moderation coefficient. Fourth, the systematic search was restricted to English-language publications; non-English studies (Chinese, Japanese, Korean, Southeast Asian) are underrepresented, potentially biasing regime distribution toward Advanced

economies. Fifth, reliability is established on a 20% double-coded subsample ($k = 47$ studies), with the two authors coding independently and agreement reported in Table 3.1; the remaining 80% is single-coded by the first author against the codebook, which, while standard for resource-bounded syntheses, cannot be independently validated.

Future research. Three directions follow directly from the null moderator findings. First, a formal WoS/Scopus systematic search targeting Frontier and SIDS economy I to P studies (targeted search strings, Appendix B) would expand Frontier k from 3 to potentially 20–40 studies, enabling a meaningful test of the ICRV Frontier cell. Second, a pre-registered replication of the ICRV, cDAI, and DPL moderator hypotheses, with explicit power analysis for each regime cell and OSF registration, would provide a definitive test of whether these moderators reach significance in a $k \geq 400$ corpus. Third, the substantial publication bias finding ($k = 57$ imputed, adj. $r = 0.035$) motivates a dedicated publication bias audit: surveying I to P researchers about unpublished null results, and meta-analyzing grey literature and dissertations, would bound the true population effect more precisely than trim-and-fill alone.

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Competing Interests

The authors declare no conflict of interest.

Data Availability

The coded study database (`forest_data.csv`), R analysis scripts (`metafor`), and the OSF registration protocol are available from the corresponding author upon reasonable request. The PRISMA 2020 checklist is provided as supplementary material.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used two software aids, each under full human control and verification. First, a purpose-built large-language-model-assisted tool (M-AIDA, using Anthropic Claude) was used to assist effect-size extraction and statistical conversion from the primary studies; every value proposed by the tool was independently checked, corrected where necessary, and permanently locked by the Principal Investigator before entry into the analysis database. Second, Grammarly was used to correct spelling, grammar, and punctuation in the authors' own text. No generative AI tool selected studies, made eligibility decisions, performed the statistical analysis, or generated any interpretive or written content; the authors reviewed all outputs and take full responsibility for the content of the publication.

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- Appendices A–C (PRISMA flow, coding protocol, MetaEssentials/metafor consistency check) are provided as online supplementary material.*